Workshop 4: solutions

1. Neither is a GLM: in both cases, the random response variable (rather than its expected value) is determined by the linear predictor. This type of model cannot be written in the standard GLM form. Note that the two models do not even make sense under the assumption of non-random predictors, as they link a random variable (or its transformation, which is still random) to something deterministic.

2. Count data suggests using the Poisson distribution, i.e., $y_i \overset{\text{ind}}{\sim} \text{Poisson}(\mu_i)$, for $i = 1, \ldots, n$. The model is
$$\mu_i \equiv \mathbb{E}(y_i) = (a + bx_i)^2,$$
and can be made linear in its parameters by employing a square root link function:
$$g(\mu_i) = \sqrt{\mu_i} = a + bx_i.$$
The linear predictor is hence of the form
$$\boldsymbol{\eta} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}.$$

3. Taking the logarithm yields
$$\log(\mu_i) = \log(n) + \log(\gamma_k) + \log(\alpha_j), \text{ if } y_i \text{ is gender } k \text{ and faith } j.$$
Defining $\tau = \log(n)$, $\beta_k = \log(\gamma_k)$, and $\delta_j = \log(\alpha_j)$, we get
$$\log(\mu_i) = \tau + \beta_k + \delta_j, \text{ if } y_i \text{ is gender } k \text{ and faith } j.$$
with $y_i \overset{\text{ind.}}{\sim} \text{Poi}(\mu_i)$ — clearly a GLM. If the data are ordered FB, FN, MB, MN (using obvious initials) then the linear predictor is
$$\begin{pmatrix} 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \tau \\ \beta_1 \\ \beta_2 \\ \delta_1 \\ \delta_2 \end{pmatrix}.$$
Obviously this is not identifiable, but by setting $\beta_1 = \delta_1 = 0$ (therefore removing columns 2 and 4 of the design matrix)
$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} \tau \\ \beta_2 \\ \delta_2 \end{pmatrix}.$$

4. A 'logistic regression' model might be appropriate (as used for the heart attack data in the notes). In particular, treating each of the six O-rings at a given launch temperature as independent binomial trials, with a failure probability dependent temperature, yields:

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right) = \beta_1 + \beta_2 \texttt{temp}_i, \quad y_i \overset{\text{ind.}}{\sim} \text{binom}(p_i, 6),$$

where p_i denotes the probability of O-ring failure or damage at temperature temp_i and y_i denotes the number of damaged O-rings (out of siz) at temperature temp_i. Thus, the model matrix would have its i^{th} row as $(1, \texttt{temp}_i)$.

5. Let us start by writing $f(y;\lambda)$ as $\exp\{\log f(y;\lambda)\}$:

$$\log f(y;\lambda) = \log\lambda - \lambda y \Rightarrow \exp\{\log f(y;\lambda)\} = \exp\{\log\lambda - \lambda y\} = \exp\left\{\frac{y(-\lambda) - (-\log\lambda)}{1} + 0\right\}.$$

This is in the exponential family of distributions with:

- $\theta = -\lambda \Rightarrow \lambda = -\theta$. Note that because $\lambda > 0$, we have that $\theta < 0$.
- $b(\theta) = -\log\lambda = -\log(-\theta)$.
- $a(\phi) = \phi = 1$.
- $c(y,\phi) = 0$.

We have established that $\mu \equiv E(Y) = b'(\theta)$, and in this case, $b'(\theta) = -\frac{1}{\theta}$.
Sanity check: The expectation of an exponentially distributed variable is known to be $\mu \equiv E(Y) = 1/\lambda$. We have derived that $E(Y) = -1/\theta$ and that $\theta = -\lambda$. Consequently, $E(Y) = 1/\lambda$, confirming consistency. Note that, in this case, the distribution was not parameterized in terms of its mean. Furthermore, we have that

$$b''(\theta) = \frac{1}{\theta^2} \Rightarrow \text{var}(Y) = b''(\theta) \times a(\phi) = \frac{1}{\theta^2} \times 1 = \frac{1}{\theta^2} = \frac{1}{\lambda^2} = \mu^2.$$

6. (a) By definition,

$$\mu \equiv \mathbb{E}[Y] = \sum_{y=0}^{1} y f(y;p) = 0 \times (1-p) + 1 \times p = p.$$

(b) We can rewrite the probability mass function of the Bernoulli distribution in terms of its mean by simply replacing p by μ. We thus have that

$$\log f(y;\mu) = y\log\mu + (1-y)\log(1-\mu),$$

leading to

$$\begin{aligned}\exp\{\log f(y;\mu)\} &= \exp\{y\log\mu + (1-y)\log(1-\mu)\}\\ &= \exp\{y\log\mu + \log(1-\mu) - y\log(1-\mu)\}\\ &= \exp\left\{y\log\left(\frac{\mu}{1-\mu}\right) + \log(1-\mu)\right\}\\ &= \exp\left\{\frac{y\log\left(\frac{\mu}{1-\mu}\right) - [-\log(1-\mu)]}{1} + 0\right\}.\end{aligned}$$

This is in the exponential family of distributions with:

- $\theta = \log\left(\frac{\mu}{1-\mu}\right) \Rightarrow \mu = \frac{e^\theta}{1+e^\theta}$.
- $b(\theta) = -\log(1-\mu) = \log(1+e^\theta)$.
- $a(\phi) = \phi = 1$.
- $c(y, \phi) = 0$.

7. We have that

$$\begin{aligned}
\exp\{\log f(y;\pi,n)\} &= \exp\left\{\log\binom{n}{ny} + ny\log\pi + (n-ny)\log(1-\pi)\right\} \\
&= \exp\left\{\log\binom{n}{ny} + ny\log\left(\frac{\pi}{1-\pi}\right) + n\log(1-\pi)\right\} \\
&= \exp\left\{\frac{y\log\left(\frac{\pi}{1-\pi}\right) - (-\log(1-\pi))}{1/n} + \log\binom{n}{ny}\right\},
\end{aligned}$$

which has the form of a density/pmf of a distribution in the exponential family with:

- $\theta = \log\left(\frac{\pi}{1-\pi}\right)$, which means that $\pi = \frac{e^\theta}{1+e^\theta}$.
- $b(\theta) = -\log(1-\pi) = \log(1+e^\theta)$.
- $a(\phi) = 1/n$.
- $c(y, \phi) = \log\binom{n}{ny}$.

8. (a) As before, let us write $f(y;\mu,\gamma) = \exp\{\log f(y;\mu,\gamma)\}$. We have that

$$\begin{aligned}
\log f(y;\mu,\gamma) &= \frac{1}{2}\log(\gamma) - \frac{1}{2}\log(2\pi y^3) - \frac{\gamma(y-\mu)^2}{2\mu^2 y} \\
&= \frac{1}{2}\log(\gamma) - \frac{1}{2}\log(2\pi y^3) - \frac{\gamma(y^2 - 2\mu y + \mu^2)}{2\mu^2 y} \\
&= \frac{1}{2}\log(\gamma) - \frac{1}{2}\log(2\pi y^3) - \frac{\gamma y}{2\mu^2} + \frac{\gamma}{\mu} - \frac{\gamma}{2y}
\end{aligned}$$

and so

$$\begin{aligned}
\exp\{\log f(y;\mu,\gamma)\} &= \exp\left\{\frac{1}{2}\log(\gamma) - \frac{1}{2}\log(2\pi y^3) - \frac{\gamma y}{2\mu^2} + \frac{\gamma}{\mu} - \frac{\gamma}{2y}\right\} \\
&= \exp\left\{\frac{y[-1/(2\mu^2)] - (-1/\mu)}{1/\gamma} + \left(\frac{1}{2}\log\gamma - \frac{1}{2}\log(2\pi y^3) - \frac{\gamma}{2y}\right)\right\}
\end{aligned}$$

This is in the exponential family of distributions with:

- $\theta = -\frac{1}{2\mu^2}$ (note that $\theta < 0$) $\Rightarrow \mu = \sqrt{-\frac{1}{2\theta}}$.
- $b(\theta) = -\frac{1}{\mu} = -\sqrt{-2\theta}$.
- $a(\phi) = \phi = \frac{1}{\gamma}$.
- $c(y,\phi) = \frac{1}{2}\log\gamma - \frac{1}{2}\log(2\pi y^3) - \frac{\gamma}{2y} = -\frac{1}{2}\log(\gamma^{-1}2\pi y^3) - \frac{\gamma}{2y} = -\frac{1}{2}\left[\log(\phi 2\pi y^3) + \frac{1}{\phi y}\right]$.

(b) From the properties of the exponential family we have that

$$\mathbb{E}(Y) = b'(\theta), \quad \text{and} \quad \text{var}(Y) = b''(\theta)a(\phi).$$

From part (a) we have that $b(\theta) = -\sqrt{-2\theta}$ and so

$$b'(\theta) = \frac{\mathrm{d}}{\mathrm{d}\theta}(-\sqrt{-2\theta}).$$

I will do the derivative step by step to avoid any mistakes. Letting $y = -\sqrt{v}$ and $v = -2\theta$, we get

$$\frac{\mathrm{d}}{\mathrm{d}\theta}(-\sqrt{-2\theta}) = \frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{\mathrm{d}y}{\mathrm{d}v}\frac{\mathrm{d}v}{\mathrm{d}\theta} = \left(\frac{\mathrm{d}}{\mathrm{d}v}(-\sqrt{v})\right)\left(\frac{\mathrm{d}}{\mathrm{d}\theta}(-2\theta)\right) = \left(-\frac{1}{2}v^{-1/2}\right)(-2) = v^{-1/2}.$$

Because $v = -2\theta$ we have

$$v^{-1/2} = (-2\theta)^{-1/2} = \frac{1}{\sqrt{-2\theta}} = \sqrt{-\frac{1}{2\theta}} = \mu.$$

So we conclude that $\mathbb{E}(Y) = \mu$. This should not come as a surprise, as it was already stated in the question text that μ is the mean.
We just need now to obtain the second derivative of $b(\theta)$. Letting $y = \frac{1}{\sqrt{v}} = v^{-1/2}$ and $v = -2\theta$ we have

$$b''(\theta) = \frac{\mathrm{d}}{\mathrm{d}\theta}b'(\theta) = \frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{\mathrm{d}y}{\mathrm{d}v}\frac{\mathrm{d}v}{\mathrm{d}\theta} = \left(\frac{\mathrm{d}}{\mathrm{d}v}v^{-1/2}\right)\left(\frac{\mathrm{d}}{\mathrm{d}\theta}(-2\theta)\right) = \left(-\frac{1}{2}v^{-3/2}\right)(-2) = v^{-3/2}.$$

Therefore,

$$v^{-3/2} = (-2\theta)^{-3/2} = \left(\sqrt{-\frac{1}{2\theta}}\right)^3 = \mu^3.$$

Finally, and recalling that in part (a) we have found that $a(\phi) = 1/\gamma$, we have

$$\text{var}(Y) = b''(\theta)a(\phi) = \frac{\mu^3}{\gamma}.$$

(c) The canonical link function is such that $g(\mu) = \theta$. Therefore, the canonical link function is

$$g(\mu) = -\frac{1}{2\mu^2}.$$

9. We have that

$$\exp\{\log f(y;\nu)\} = \exp\left\{\log\Gamma\left(\frac{\nu+1}{2}\right) - \frac{1}{2}\log(\nu\pi) - \log\Gamma\left(\frac{\nu}{2}\right) - \frac{\nu+1}{2}\log\left(1+\frac{y^2}{\nu}\right)\right\}$$

As we can see, this expression does not simplify into a form where there is a linear term in y corresponding to $y\theta$ in the exponential family of distributions representation. Thus, the t-distribution does not belong to the exponential family.

10. Note that $Y_i \in \{0, 1, 2, 3, \dots\}$ and, as such,

$$Z_i = \begin{cases} 1, & \text{if } Y_i > 0, \\ 0, & \text{if } Y_i = 0, \end{cases}$$

and hence it holds that

$$\begin{aligned} \pi_i &= \Pr(Z_i = 1) \\ &= 1 - \Pr(Z_i = 0) \\ &= 1 - \Pr(Y_i = 0). \end{aligned}$$

Because $Y_i \overset{\text{ind}}{\sim} \text{Poisson}(\mu_i)$, we know that

$$\Pr(Y_i = 0) = \frac{e^{-\mu_i}\mu_i^0}{0!} = e^{-\mu_i},$$

and hence

$$\pi_i = 1 - e^{-\mu_i}.$$

Solving for μ_i we get

$$1 - \pi_i = e^{-\mu_i} \Rightarrow \mu_i = -\log(1 - \pi_i).$$

Because $\log(\mu_i) = \mathbf{x}_i^\mathsf{T}\boldsymbol{\beta}$, we have that

$$\log(\mu_i) = \log(-\log(1 - \pi_i)) = \mathbf{x}_i^\mathsf{T}\boldsymbol{\beta}.$$

Note that $\mathbb{E}(Z_i) = 1 \times \Pr(Z_i = 1) + 0 \times \Pr(Z_i = 0) = \Pr(Z_i = 1) = \pi_i$. Out of curiosity, this link function is known as the complementary log-log link (covered in more detail in Workshop 5).

11. Let us start by calculating the expected value of the vector $\mathbf{z}$. We have that

$$\begin{aligned} \mathbb{E}(z_i) &= \mathbb{E}[g'(\mu_i)(y_i - \mu_i) + \mathbf{x}_i^\mathsf{T}\boldsymbol{\beta}] \\ &= g'(\mu_i)[\mathbb{E}(y_i) - \mu_i] + \mathbf{x}_i^\mathsf{T}\boldsymbol{\beta} \\ &= g'(\mu_i)(\mu_i - \mu_i) + \mathbf{x}_i^\mathsf{T}\boldsymbol{\beta} \\ &= \mathbf{x}_i^\mathsf{T}\boldsymbol{\beta}, \end{aligned}$$

and so $\mathbb{E}(\mathbf{z}) = \mathbf{X}\boldsymbol{\beta}$. From standard results on variance transformations, we have that

$$\begin{aligned} \text{var}(z_i) &= \text{var}(g'(\mu_i)(y_i - \mu_i) + \mathbf{x}_i^\mathsf{T}\boldsymbol{\beta}) \\ &= \text{var}(g'(\mu_i)y_i - g'(\mu_i)\mu_i + \mathbf{x}_i^\mathsf{T}\boldsymbol{\beta}) \\ &= g'(\mu_i)^2\text{var}(y_i) \\ &= g'(\mu_i)^2 V(\mu_i)\phi. \end{aligned}$$

Since the y_i are mutually independent, so are the z_i. Hence the covariance matrix for $\mathbf{z}$ is a diagonal matrix whose ith entry is given by $g'(\mu_i)^2 V(\mu_i)\phi$. Using the notation from the lecture notes (section 11.2), $g'(\mu_i)^2 V(\mu_i) = \mathrm{w}_i^{-1}$, and hence we can also write the covariance matrix for $\mathbf{z}$ as $\mathbf{W}^{-1}\phi$ where $\mathbf{W}$ is a diagonal matrix, with $W_{ii} = \mathrm{w}_i$.

12. (a)
$$\frac{\mathrm{d}}{\mathrm{d}\beta}\sum_{i=1}^{n}\mathrm{w}_i(z_i-\beta x_i)^2=-2\sum_i \mathrm{w}_i(z_i-x_i\beta)x_i$$
Setting this to zero and re-arranging yields:
$$\hat{\beta}=\frac{\sum_i \mathrm{w}_i z_i x_i}{\sum_i \mathrm{w}_i x_i^2}.$$
Differentiating the weighted sum of squares again shows $\hat{\beta}$ *minimizes* it.

(b) The link function is the identity function, $g(\mu_i)=\mu_i$, and so $g'(\mu_i)=1$. For the Poisson distribution we have $V(\mu_i)=\mu_i$ (see Section 11.1 of the notes). Hence, at each IRLS iteration we will have
$$\mathrm{w}_i=(g'(\mu_i)^2V(\mu_i))^{-1}=\frac{1}{\mu_i},$$
and
$$z_i=g'(\mu_i)(y_i-\mu_i)+x_i\beta=(y_i-\mu_i)+x_i\beta=(y_i-x_i\beta)+x_i\beta=y_i.$$
Note that $g(\mu_i)=\eta_i=x_i\beta$, but because g is the identity function, $g(\mu_i)=\mu_i=\eta_i=x_i\beta$.
So the IRLS steps will proceed as follows.

- **Initialize** Set $\boldsymbol{\mu}^{(0)}=(1,1,3,7,4)^\mathsf{T}$ (i.e. set $\boldsymbol{\mu}^{(0)}$ to $\mathbf{y}$, except when $y_i=0$, since μ_i needs to be greater than zero). $\boldsymbol{\eta}^{(0)}=\boldsymbol{\mu}^{(0)}$, since the model has an identity link.
- **Iteration 1** First find the iterative weights and working/pseudo responses:
$$\mathrm{w}_i^{(0)}=(\mu_i^{(0)})^{-1}\Rightarrow(\mathrm{w}_1^{(0)},\mathrm{w}_2^{(0)},\mathrm{w}_3^{(0)},\mathrm{w}_4^{(0)},\mathrm{w}_5^{(0)})=\left(1,1,\frac{1}{3},\frac{1}{7},\frac{1}{4}\right),$$
$$z_i^{(0)}=y_i\Rightarrow(z_1^{(0)},z_2^{(0)},z_3^{(0)},z_4^{(0)},z_5^{(0)})=(0,1,3,7,4).$$
Using part (a) we have that $\beta^{(1)}=\frac{\sum_{i=1}^{5}\mathrm{w}_i^{(0)}z_i^{(0)}x_i}{\sum_i \mathrm{w}_i^{(0)}x_i^2}$. I will do this calculation in R.

```
omega_0 <- c(1, 1, 1/3, 1/7, 1/4)
z_0 <- c(0, 1, 3, 7, 4)
x <- c(1, 2, 3, 4, 5)
beta_1 <- sum(omega_0*z_0*x)/sum(omega_0*(x^2))
beta_1

## [1] 0.8466523
```

- **Iteration 2** We have that $\mathrm{w}_i^{(1)}=(\mu_i^{(1)})^{-1}=(x_i\beta^{(1)})^{-1}$ and $z_i^{(1)}=y_i$. We can then calculate $\beta^{(2)}$ as in the first iteration.

```
mu_1<- x*beta_1
omega_1 <- 1/mu_1
z_1 <- c(0, 1, 3, 7, 4)
beta_2 <- sum(omega_1*z_1*x)/sum(omega_1*(x^2))
beta_2

## [1] 1
```

Actually, $\hat{\beta}=1$ is the MLE — in this case the IRLS has converged in 2 steps. Let us confirm this.

```r
x <- c(1, 2, 3, 4, 5)
y <- c(0, 1, 3, 7, 4)
res_toy <- glm(y ~ x-1, family = poisson(link = "identity"))
res_toy$coefficients

## x
## 1
```

(c) From general theory, we know that (asymptotic distribution of the maximum likelihood estimate $\hat{\boldsymbol{\beta}}$) the covariance matrix for a GLM with scale parameter (ϕ) equal to 1 is $(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$. For the current single parameter model, this becomes $\left(\sum_i \mathrm{w}_i x_i^2\right)^{-1}$, with the square root of this giving the standard error for $\hat{\beta}$. To evaluate it we need the final estimates for the w_i:

$\hat{\mu}_i = \hat{\eta}_i = x_i\hat{\beta} = x_i$	1	2	3	4	5
$\mathrm{w}_i = \hat{\mu}_i^{-1}$	1	1/2	1/3	1/4	1/5

which results in a standard error estimate for $\hat{\beta}$ of 0.258. We can check this is indeed the correct value by comparing to the value returned by the `glm` function.

```r
sqrt(vcov(res_toy))

##           x
## x 0.2581989
```